

Report

Haensel AMS Recruitment Challenge

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1 How did I model spend carry over?

I used adstock transformation to account for carry over using the recursive decay formula :
 $\text{adstock}[t] = \text{spend}[t] + \alpha \times \text{adstock}[t-1]$

α is the decay rate, constrained between 0 and 1.

Fast decay (α near 0) meaning effect is immediate with no carry-over. Slow decay (α near 1) meaning the effect lingers for a longer time.

Here Beta(2,2) prior centered at 0.5 was used (no strong assumption about whether decay would be fast or slow, and allowing the data to decide).

Alpha was estimated independently for each channel rather than fixed as a constant. The posterior estimates ranged from 0.25 to 0.42 across all seven channels, suggesting fast-to-moderate decay throughout.

2 Explaining my choice of prior inputs to the model

The prior inputs were chosen according to the problem

1. Intercept : Normal(0.3, 0.1)

Centered at 0.3 on scaled revenue to account for the baseline revenue even without any marketing activity .

A sigma of 0.5 was initially set which proved to be too wide ,it was then tightened to 0.1 after prior predictive checking to prevent the intercept from dominating the prediction range.

2. Channel coefficients (beta_channel) :HalfNormal(sigma=0.14)

HalfNormal was chosen because marketing spend cannot causally destroy revenue (negative revenue values) , only 0 or positive values makes sense.

Initially a spend-proportional sigma was used, where channels with larger budgets received wider priors. But, the prior predictive check revealed that the generated revenue predictions exceeded €8M against an observed maximum of €418k. Then a uniform sigma of 0.14 was iteratively selected until the prior predictive value fell within a plausible range.

3. Adstock decay (alpha) : Beta(2,2)

Prior centered at 0.5 was used ,no strong assumption of whether decay would be fast or slow, allowing data to decide.

4. Trend (beta_trend) :Normal(0, 0.15)

Normal was chosen to allow the trend go positive or negative since the business direction over the period was unknown at modelling time.

5. Fourier seasonality (γ_{fourier}) : $\text{Laplace}(0, 0.2)$

Laplace was chosen over Normal because it keeps most coefficients close to zero by default, only allowing seasonal terms to grow large if the data clearly supports it.

This felt like the safer choice as there was only two years of data (August 2020 to August 2022) using a more flexible prior might make the model fit random weekly noise as if it were a seasonal pattern.

6. Observation noise (σ_{obs}) : $\text{HalfNormal}(0.1)$

HalfNormal as it must be positive by definition. This was also later tightened after checking the prior predictive values, it prevents the model from hiding poor fit behind a large noise term.

In short, my choice of prior values was iterative, it was adjusted until the prior predictive revenue fell within a plausible range

3 How are my model results based on prior sampling vs. posterior sampling?

Prior predictive sampling was used to check if the priors I chose made sense before fitting the model. With 500 prior samples, the model generated weekly revenue between €84k–€656k (10th–90th percentile), which looked reasonable given the observed data.

After fitting the model, posterior predictive sampling shows how well the model explains the actual revenue.

The main difference between prior and posterior is that prior samples are our initial assumptions and posterior samples are what the model actually learned from the data. The posterior distributions are noticeably narrower than the prior, so we can say that the data was informative and the model picked up real patterns.

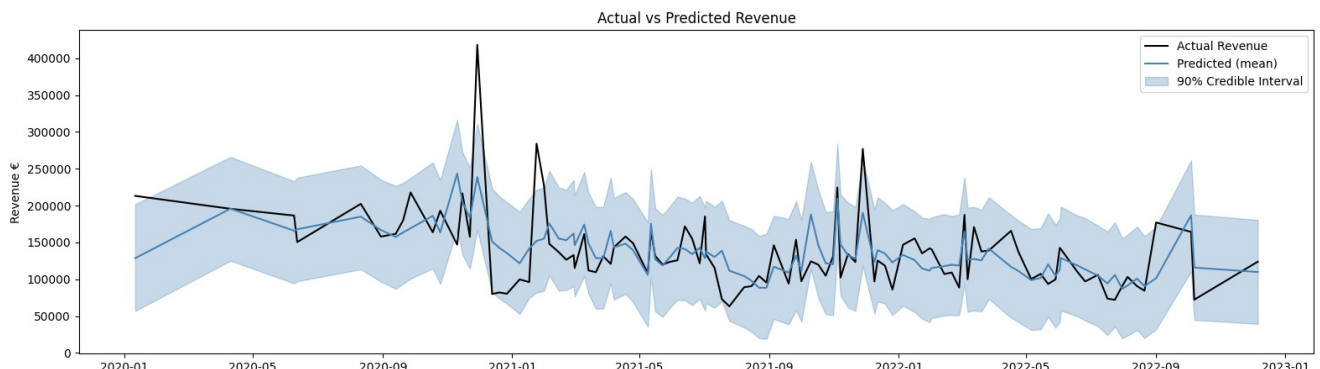
4 How good is my model performing? How did I measure it?

I measured the model performance in two ways : convergence and fit.

For convergence, $\hat{r}$ of 1.0 (<1.01 recommended) and minimum ESS of 1059 (>400 recommended) was observed which confirm the model sampled correctly and the results are reliable. (Convergence thresholds follow Vehtari et al. (2021))

For fit, the model achieves an R^2 of 0.42 (explains 42% of the variation in weekly revenue). MAPE of 20.6% means predictions are on average 20% off, and RMSE was €38,507 gives the average error in euro terms.

The actual vs predicted plot shows the model captures the general trend and most weeks fall within the 90% credible interval. Overall the model fit is moderate and reasonable.



5 What are my main insights in terms of channel performance/effects?

All 7 channels show a positive effect on revenue with $p(>0)=1.0$, meaning the model is confident every channel contributes positively.

In terms of effect size, channels 6 and 7 have the highest beta coefficients (0.106 and 0.110), meaning they have the strongest direct impact on revenue per unit of spend

In case of adstock, channels 1 and 2 have the highest alpha values (0.42 and 0.41), meaning their effects linger longest. Channel 3 has the lowest alpha meaning its effect fades the quickest.

The trend coefficient is negative (-0.131), confirming the baseline revenue declined over the period.

The Fourier seasonality coefficients are all close to zero, suggesting there is no strong seasonal pattern in this dataset over the two year period.

6 What are the ROI (return on investment) estimates per channel? What is the best channel in terms of ROI?

ROI was estimated by dividing each channel's total attributed revenue by its total spend. Attributed revenue was calculated using the posterior mean beta and adstock values per channel.

Channel 2 shows the highest ROI, however this is unreliable due to its very low total spend of €35k. A small spend with any attributed revenue gives a large ratio, so this result is not a genuine signal (ROAS very wide, so there is uncertainty in the estimate)

Setting channel 2 aside, channel 6 and 1 have very close ROI but channel 6 can be considered the best performing channel as its ROAS posterior distribution is more concentrated (channel 1 ROAS is very wide, therefore ROI unreliable). Channel 4 has the next best ROI and a better concentrated ROAS distribution than channel 1.

Channels 3 and 7 had the largest budgets, yet channel 3 barely breaks even and channel 7 is below break even point. This suggests these two channels are currently under-performing relative to their spend.

It is very important to note that since the model doesn't account for saturation effect, these ROI estimates alone cannot be used to recommend a budget reallocation.

Channel	ROI (€ per €1 spent)	Verdict
Channel 6	1.99	Best channel (good ROI and reliable ROAS)
Channel 4	1.83	Good ROI and more reliable than channel 1
Channel 5	1.28	Marginal but above break-even
Channel 3	1.09	Barely breaks even despite high spend
Channel 1	1.93	Good ROI but uncertain ROAS
Channel 7	0.83	Below break even (despite highest spend)
Channel 2	33.85	Highest ROI but unreliable ROAS and too low spend to conclude

Table 1: Channel ROI estimates with reliability assessment

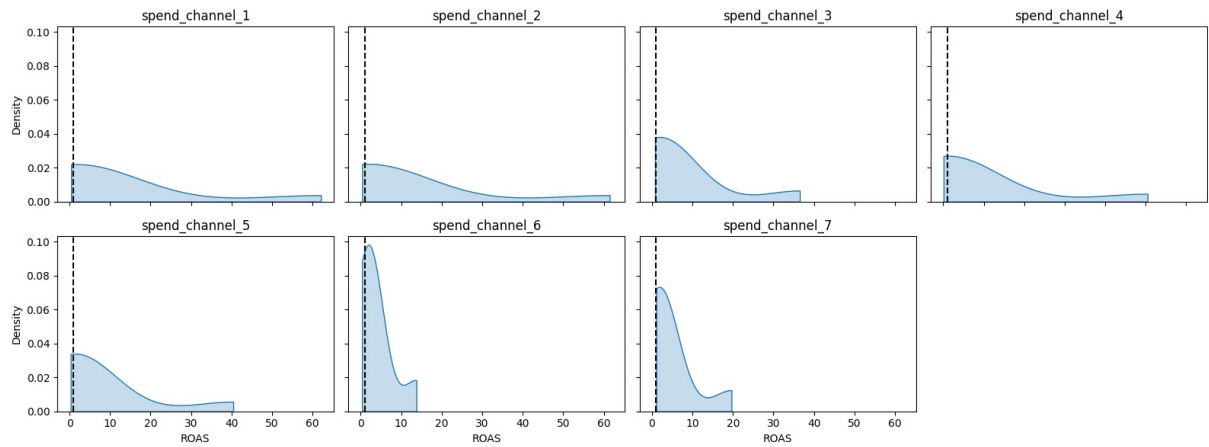


Figure 1: ROAS Posterior Distributions by Channel

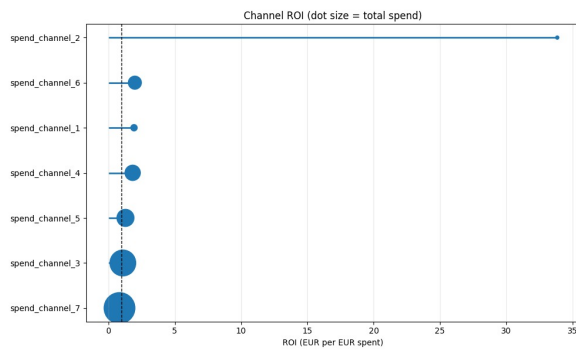


Figure 2: ROI vs spend lollipop plot

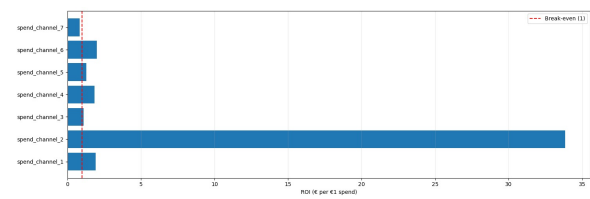


Figure 3: Spend channels ROI

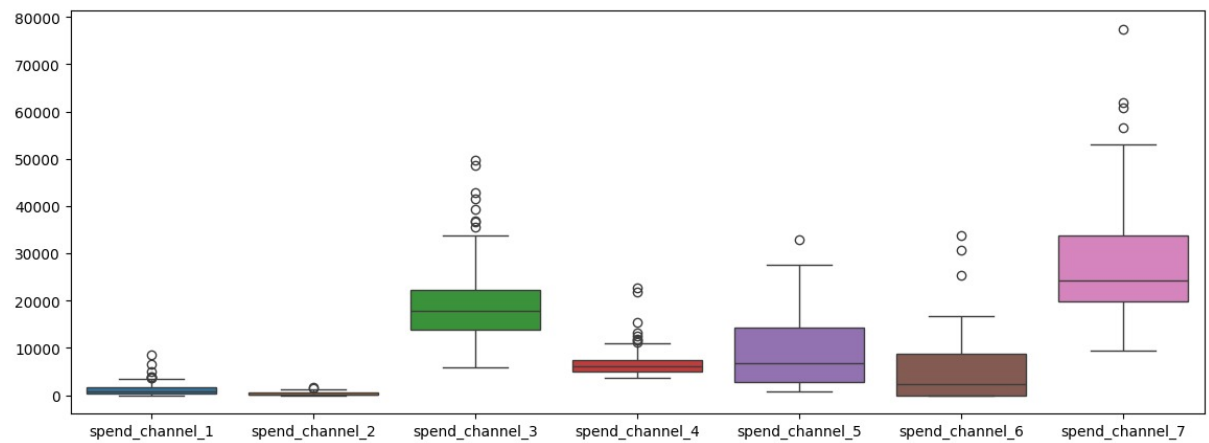


Figure 4: Box plots of all spend channels